

Hardware:

3D Printed Components:

The STL files for each part are available, as well as the .f3d file for the drone overall. The names for each component are mostly self-descriptive, with the one exception being the “Drone Pin 1.0” and “Drone Pin Long 1.0”. The finalized design mainly uses the long pin variant, but the shorter pin can be used at the cost of potential rigidity. Furthermore, the pins holding the arms to the main body can be very difficult to remove and not all the pin slots are through holes. The first camera-side pin slot on both sides of the drone is a blind hole, so caution should be taken when using the pins on those slots. This problem can be worked around by either not fully inserting the pins on the blind holes, or by designing a new pin with more material on the top for easier removal.

Propeller Blades:

The propeller blades used in this project are two-bladed 12 x 6 propellers. The drone currently has two counterclockwise propellers with these dimensions, so the next step would be to purchase two more of these propellers, with care taken to make sure that they are clockwise (the drone requires two counterclockwise propellers and two clockwise propellers to balance out torques for yaw control).

Electronics:

Battery:

The drone uses a 11.1V, 3S1P, 5000 mAh battery with a XT60 connector. This battery is compatible with the OVONIC battery charger provided.

Electronic Speed Controller (ESC):

Four 30A RCElectricParts electronic speed controllers are used. These are necessary for the controlled operation of each motor. These ESCs have three groups of connections. The XT60 female connector is used to power the ESC from the battery. The DuPont wire connectors are used to control the ESC from the circuit board, with the white wire being data, the black wire being ground, and the red wire being a 5V BEC (not used). Lastly, the three wires with “banana” connectors are used to connect to the motor, with the roles being as follows: the blue wire is the data cable and corresponds to the yellow wire on the motors, and the red and black wires are for power and ground, respectively. These wires can connect to the red and black of the motors, the

order of which changes the direction the motor spins (i.e. changing from black-black and red-red to black-red and red-black).

Motor:

Four A2208 / 8T, 2600 KV motors are used. These motors are mounted to the frame using four metric M3 bolts per motor. Each motor is exclusively powered by the ESC and should not be powered by the circuit board. The direction of each motor can be changed by swapping the ground and power cables that connect to the ESC. The drone should have two counterclockwise motors and two clockwise motors, corresponding to the type of propeller that each motor has mounted on it. The propellers are mounted onto the drone by the prop adapter, which should be tightened down thoroughly before use. **Do not** test the motors with propellers attached unless proper precautions are taken beforehand. It should be noted that the entire external body of the motor spins, not just the spigot.

Universal Battery Eliminator Circuit (UBEC):

The UBEC is used to step down the 11.1V from the battery to 5V/3A that is usable by the Raspberry Pi. This UBEC is connected to a female XT60 connector via heat shrink tubing, so avoid disconnecting the UBEC by pulling on the wires themselves. Instead, try to only pull on the exposed plastic sections of the XT60 connectors.

5-Way Splitter:

The splitter is used to split the battery's power to five different components. The splitter has one female XT60 connector and five male XT60 connectors. The female connector connects to the battery, and the male connectors connect to the four ESCs and the UBEC.

MPU-6050 6-axis Accelerometer and Gyroscope:

This gyroscopic sensor can be used to detect the orientation of the drone, as well as the acceleration in X, Y, and Z. The MPU-6050 has connectors that must be soldered with the provided adaptor to ensure the reliability of the connection during flight. These connectors should be connected to the Raspberry Pi to allow the readings to be measured. This sensor uses metric M2.5 bolts.

PCA-9685 PWM Driver:

The PWM driver is used to control all four motors at the same time. Each motor requires a PWM signal to discern analog speeds, and the Raspberry Pi only has two discrete PWM signals, so the PWM driver is necessary to control all four motors. The PWM board has 16 channels, with three pins used for each channel. The yellow pin is the signal pin which connects to the white wire of the ESC. The black pin is the ground and connects to the black wire of the ESC. Lastly, the red pin is the power pin, which would be used to power the board from the

ESCs, but the PWM driver board is being powered by the Raspberry Pi, so this pin should remain unconnected. This board uses metric M2.5 bolts.

ArduCam V3:

Connects to Raspberry Pi via CSI-2 serial port. Has adjustable framerate and video quality. Currently is mounted inverted. Attaches via ~1.5-2M bolts.

Raspberry Pi 4:

The circuit board that controls the function of the drone. More detail on the control can be found in the “Software” section. The Raspberry Pi uses a 64 GB microSD card as memory, and this card is necessary for the operation of the board. If more memory is needed, the microSD card can be changed. The board is powered either by UBEC into pins 4 and 6, or by USB-C. The board can be interfaced with by plugging a monitor into the micro-HDMI port, and a keyboard and mouse into the USB-3 ports.

The board has multiple pinouts of various purpose, including powering and providing inputs for the PCA-9685, as well as reading outputs from the MPU-6050. The Raspberry Pi directly connects to the ArduCam via the data ribbon. Additional account information found below:

Username: lpan

Password: Steven27

IP: 192.168.1.165

Software:

Raspberry Pi Wi-Fi Hotspot:

The Raspberry Pi can emit a Wi-Fi hotspot under the name of PiCam by running the following line in the Raspberry Pi terminal.

```
nmcli connection up pi-hotspot
```

The hotspot can be disabled by running the following line.

```
nmcli connection down pi-hotspot
```

Raspberry Pi Wi-Fi Video Streaming:

Using GStreamer, video feed can be streamed over the Raspberry Pi’s hotspot to a receiving computer (must be connected to the hotspot). The stream can be started with the following line:

```
Libcamera-vid -t 0 --width 640 --height --framerate 25 --  
codech264 --inline --listen -o - | gst-launch-1.0 -v fdsrc !
```

```
h264parse ! rtph264pay config-interval=1 pt=96 ! udpsink  
host=192.168.4.66 port=5000
```

Recommended First Steps:

Hardware:

- Purchase proper sized and orientation propeller blades
- Design attachment for MPU-6050

Electronics:

- Connect MPU-6050 to proper pins

Software:

- Digitally flip video camera feed
- Get readings from MPU-6050
- Run basic flight test to ensure weights and power is appropriate